Project Scheduling

Once cost and effort required are estimated, then manager make necessary planning for developing individual modules.

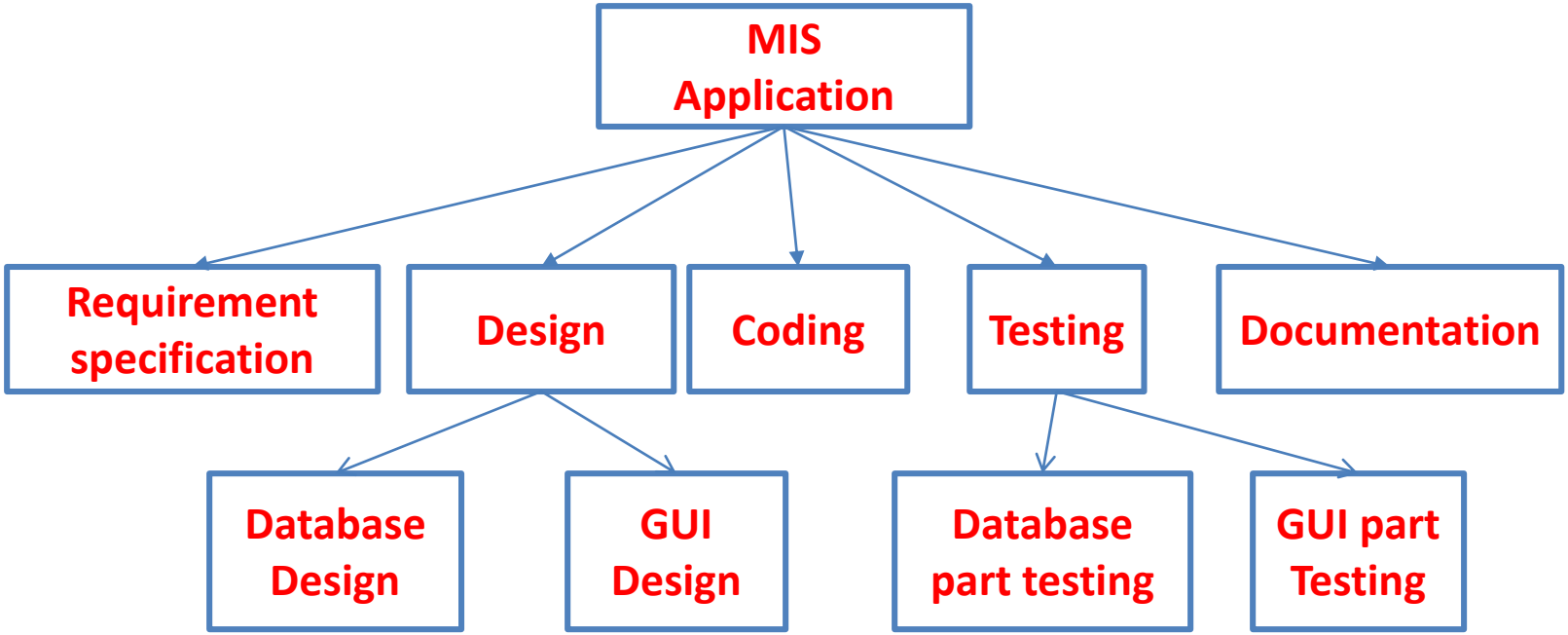
S. S. Satapathy

Project scheduling

- Project-task scheduling involves deciding which tasks would be taken up when. For project scheduling, a software project manager needs to do the following:
  - **Identify all the tasks** needed to complete the project.
  - **Break down large tasks into small activities**.
    - **The work breakdown structure formalism helps the manager to breakdown the tasks systematically**
  - **Determine** the **dependency among different activities**.
    - **Dependency among the different activities determines the order in which the different activities would be carried out. In general, the task dependencies define a partial ordering among tasks, i.e. each tasks may precede a subset of other tasks, but some tasks might not have any precedence ordering defined between them (called concurrent task). The dependency among the activities are represented in the form of an activity network.**
  - **Establish** the **most likely estimates for the time durations necessary** to complete the activities.
  - **Allocate resources to activities**.
    - **Resource allocation is typically done using a Gantt chart.**
  - **Plan the starting and ending dates for various activities**.
    - **After resource allocation is done, a PERT chart representation is developed. The PERT chart representation is suitable for program monitoring and control.**
  - **Determine the critical path. A critical path is the chain of activities that determines the duration of the project**.

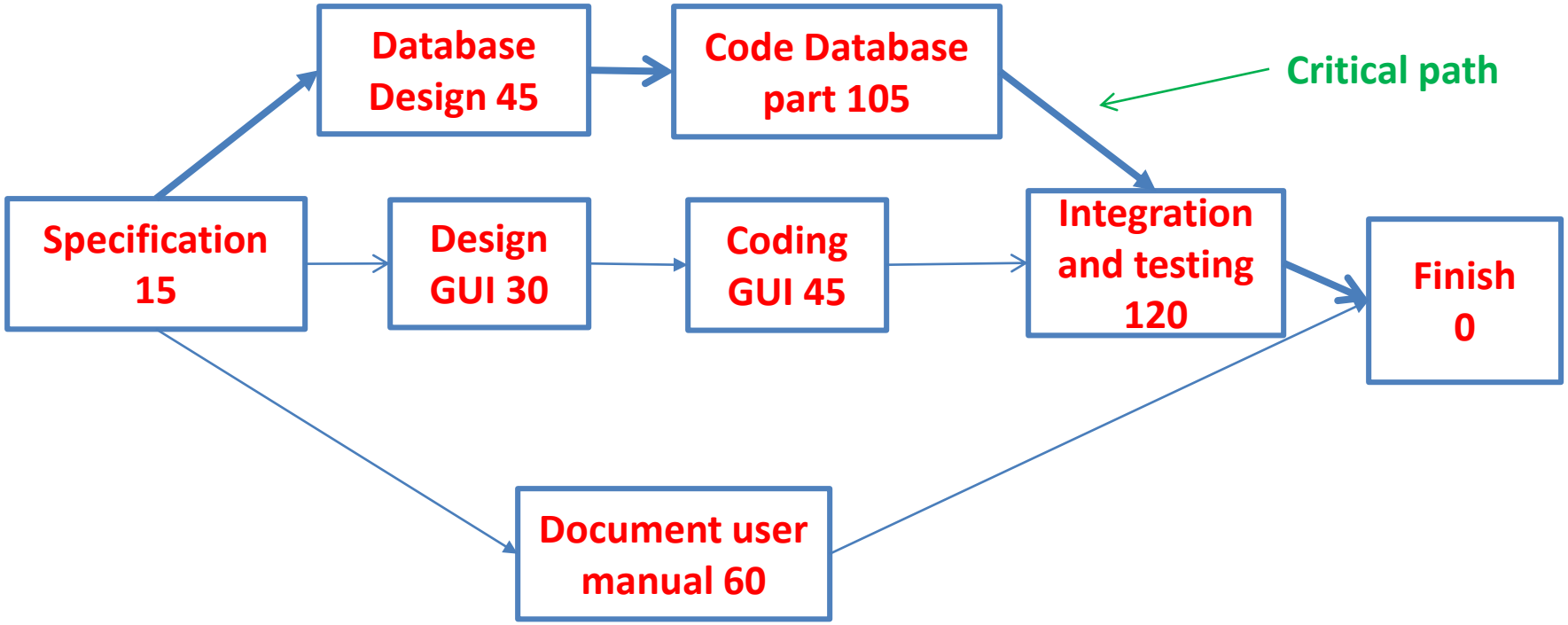
Work breakdown structure

- **Work Breakdown Structure (WBS)** is used to **decompose** a given **task** set recursively **into small activities**.
  - WBS provides a notation for representing the major tasks need to be carried out in order to solve a problem.
  - The root of the tree is labeled by the problem name.
  - Each node of the tree is broken down into smaller activities that are made the children of the node. **Each activity is recursively decomposed into smaller sub-activities until at the leaf level, the activities requires approximately two weeks to develop**.



Activity networks and critical path method

- WBS representation of a project is transformed into an activity network by representing activities identified in WBS along with their interdependencies.
  - An activity network shows the different activities making up a project, their estimated durations, and interdependencies.
  - **Each activity is represented by a rectangular node** and the **duration of the activity** is shown alongside each task.



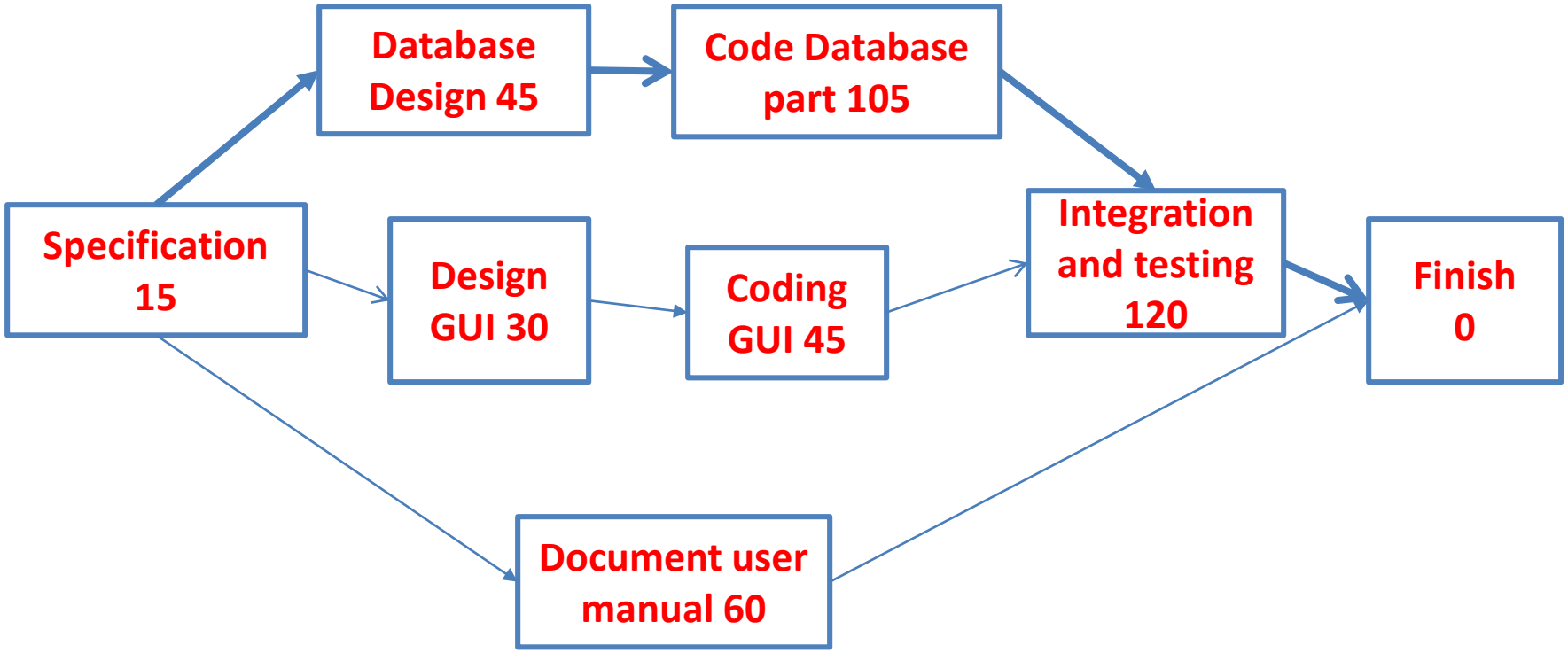
Activity network representation

Critical Path Method (CPM)

- From the activity network representation following analysis can be made
  - **The minimum time (MT) to complete the project**
  - **The earliest start (ES) time of a task**
  - **The latest start (LS) time of a task**
  - **The earliest finish time (EF) of a task**
  - **The latest finish (LF) time of a task**
  - **The slack time (ST) of a task**
  - **The critical task**
  - **The critical path**

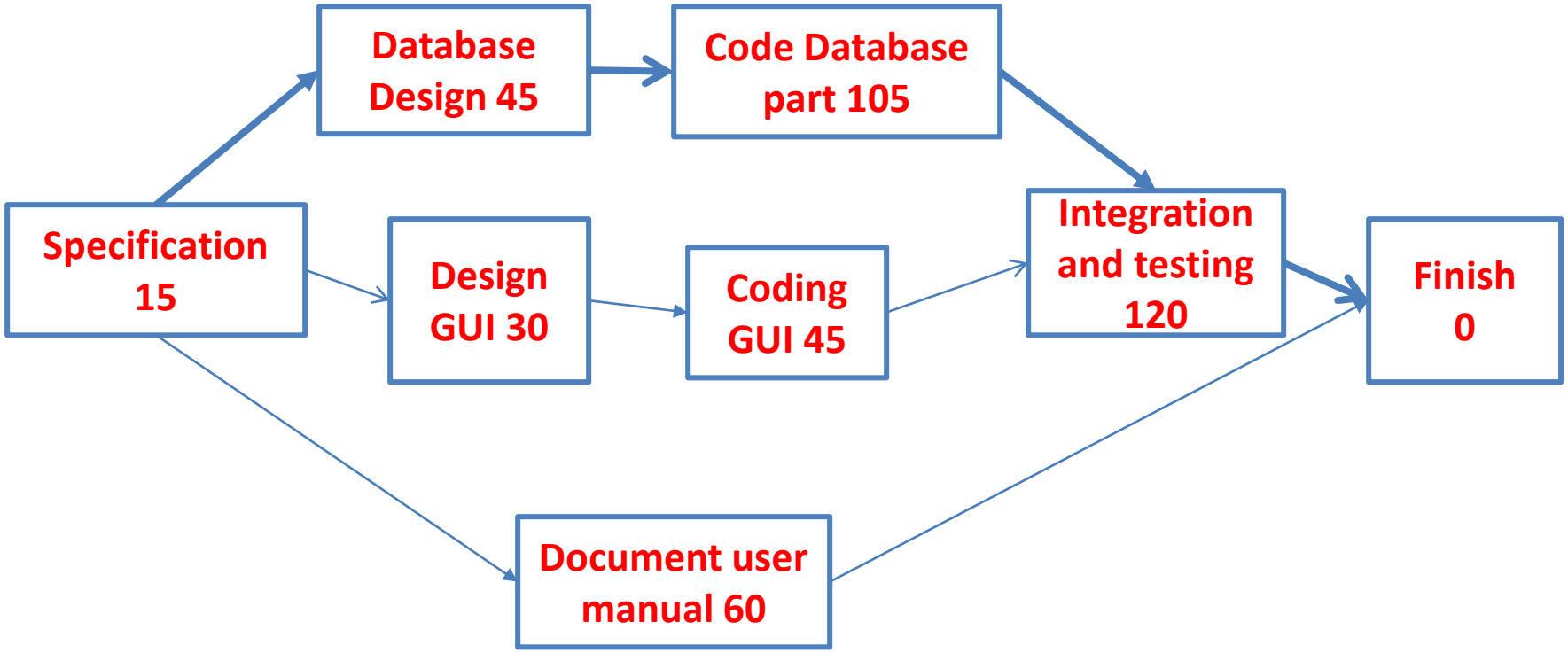
Critical Path Method (CPM)

- What is **the minimum time (MT)** to complete the project ?



Critical Path Method (CPM)

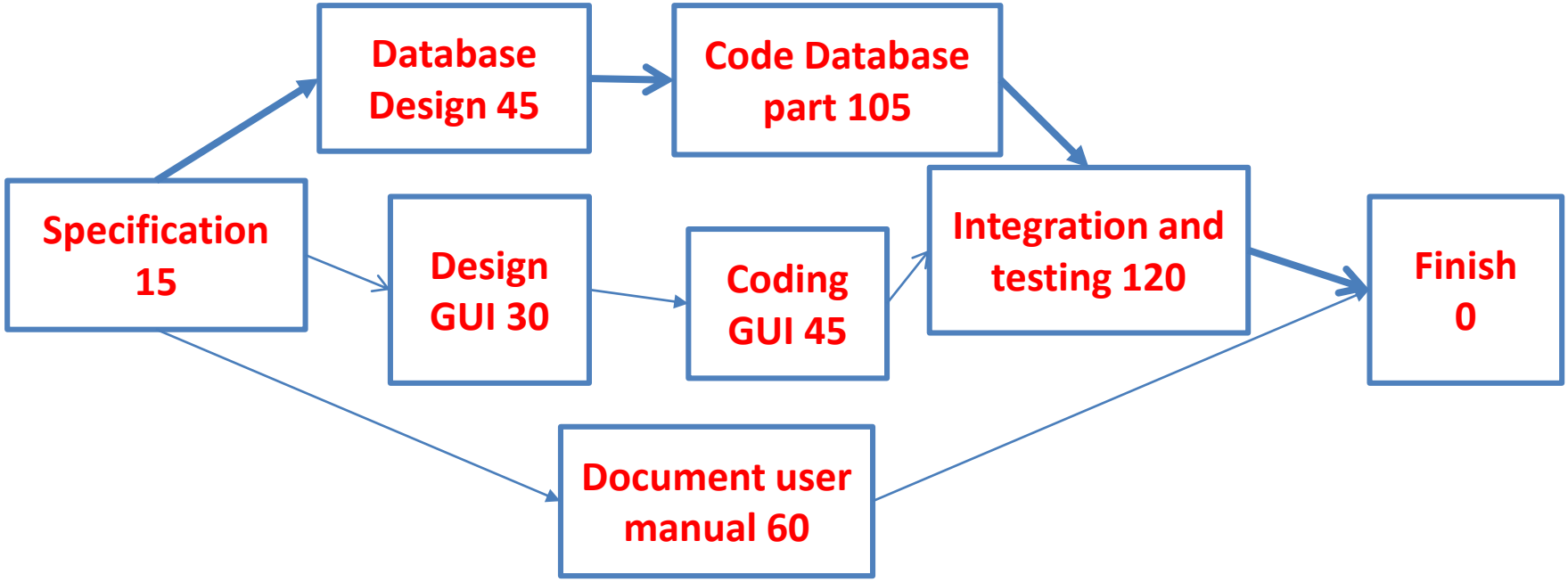
- **The minimum time (MT)** to complete the project is **the maximum of all paths from start to finish (Length of Longest path)**.
  - For this project **MT** is **15+45+105+120=285**



Critical Path Method (CPM)

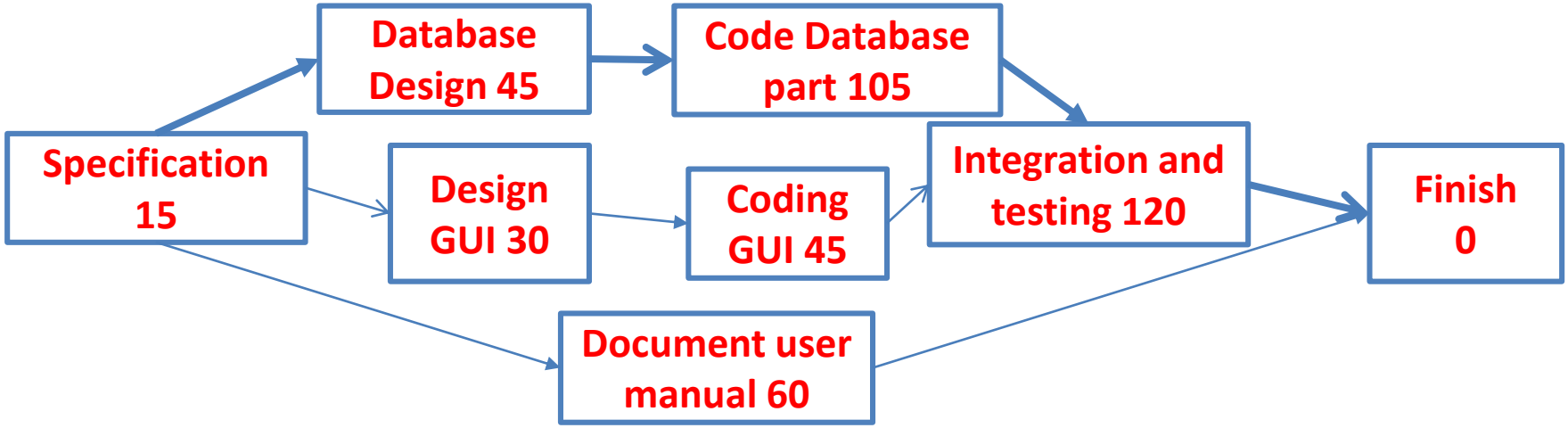
- What are the **earliest start (ES)** time of a task?

Task	Specification	Design database	Design GUI part	Code database	Code GUI part	Integrate and test	Write user manual
ES	?	?	?	?	?	?	?



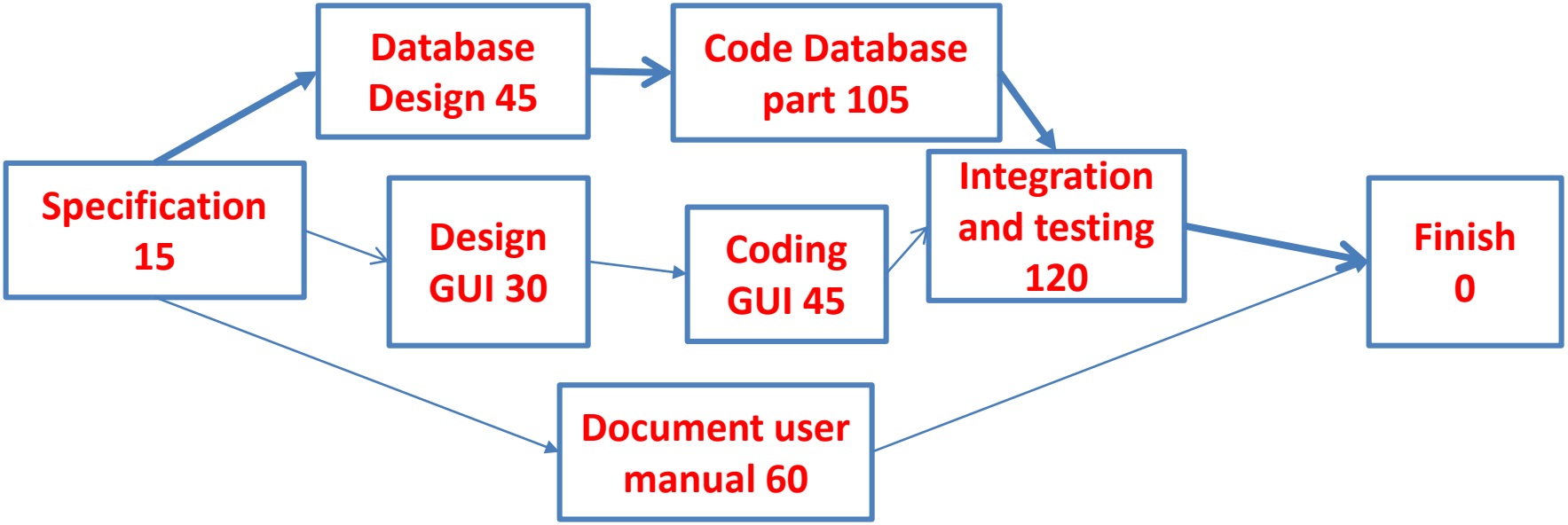
Critical Path Method (CPM)

- What are the **earliest start (ES) time of a task?**



- The **earliest start (ES) time of a task** is the **maximum of all paths from the start to the task.**
  - For Example: **ES** for **Int. and testing** is: **max(165,90)=165**

Critical Path Method (CPM)

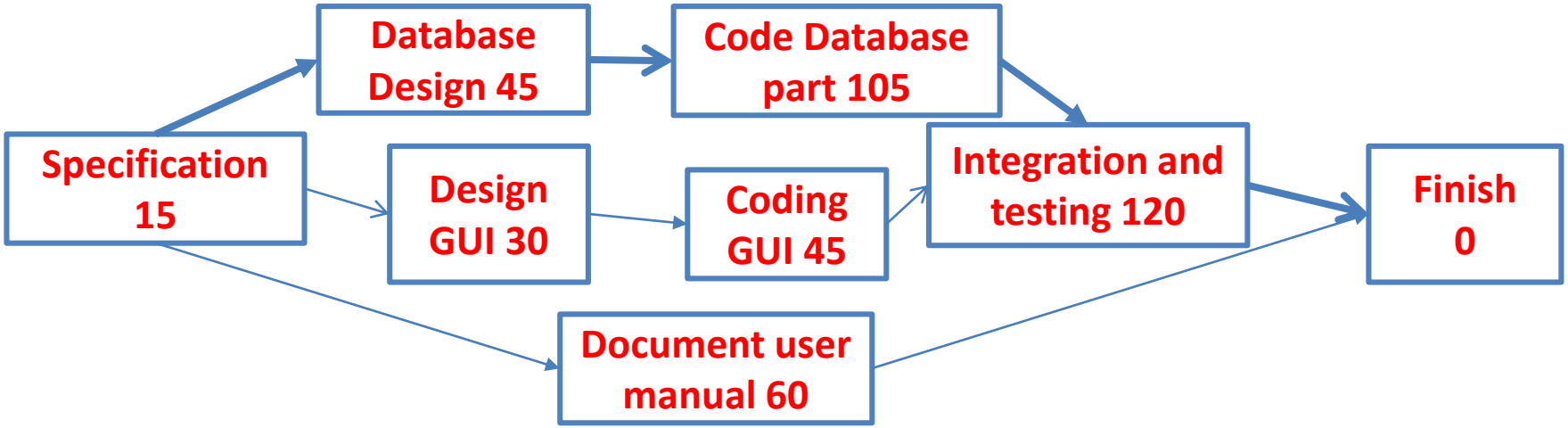


Task	Specification	Design database	Design GUI part	Code database	Code GUI part	Integrate and test	Write user manual
ES	0	15	15	60	45	165	15

Critical Path Method (CPM)

- What are the **earliest finish time (EF) of a task** ?

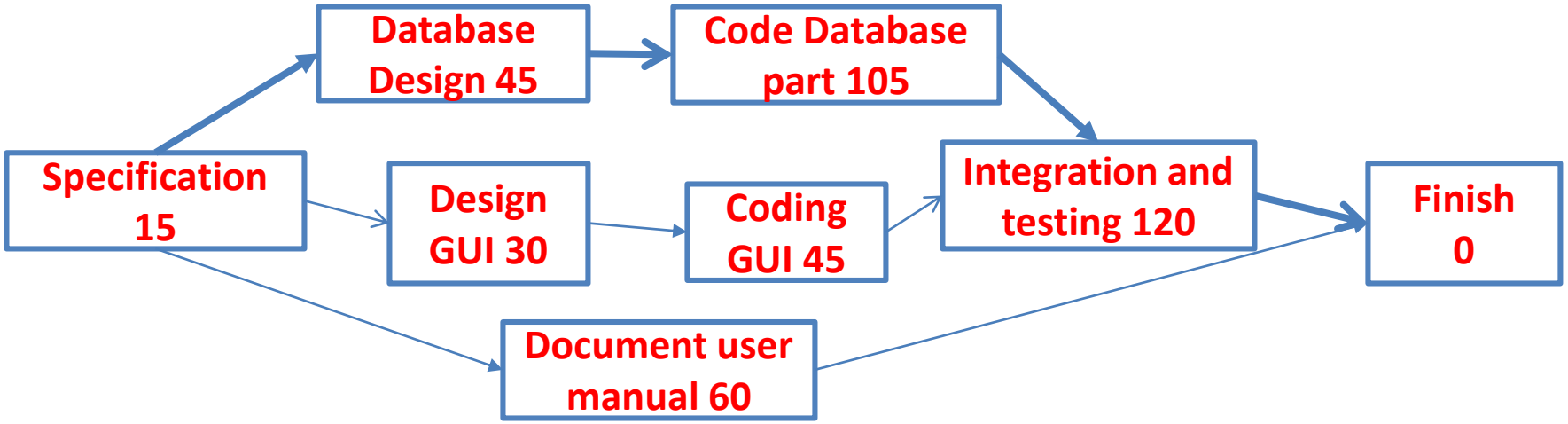
Task	Specification	Design database	Design GUI part	Code database	Code GUI part	Integrate and test	Write user manual
ES	0	15	15	60	45	165	15
EF	?	?	?	?	?	?	?



- The **earliest finish time (EF) of a task** is the **sum of the earliest start time of the task and the duration of the task (ES + task duration)**.
  - For Example: **EF** for **specification** is: **0+15=15**

Critical Path Method (CPM)

- The **earliest finish time (EF) = ES + task duration**.
  - For Example: **EF** for **specification** is: **0+15=15**

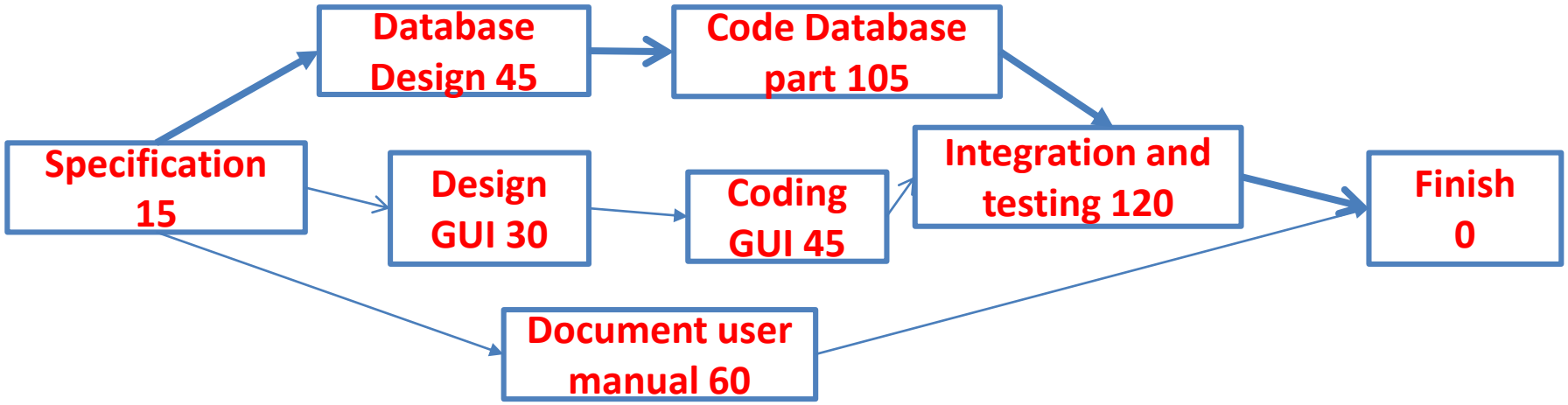


Task	Specification	Design database	Design GUI part	Code database	Code GUI part	Integrate and test	Write user manual
Duration	15	45	30	105	45	120	60
ES	0	15	15	60	45	165	15
EF	15	60	45	165	90	285	75

Critical Path Method (CPM)

- The **latest start (LS) time** of a **task ? Can we delay a task without delaying completion of the project?**

Task	Specification	Design database	Design GUI part	Code database	Code GUI part	Integrate and test	Write user manual
LS	?	?	?	?	?	?	?

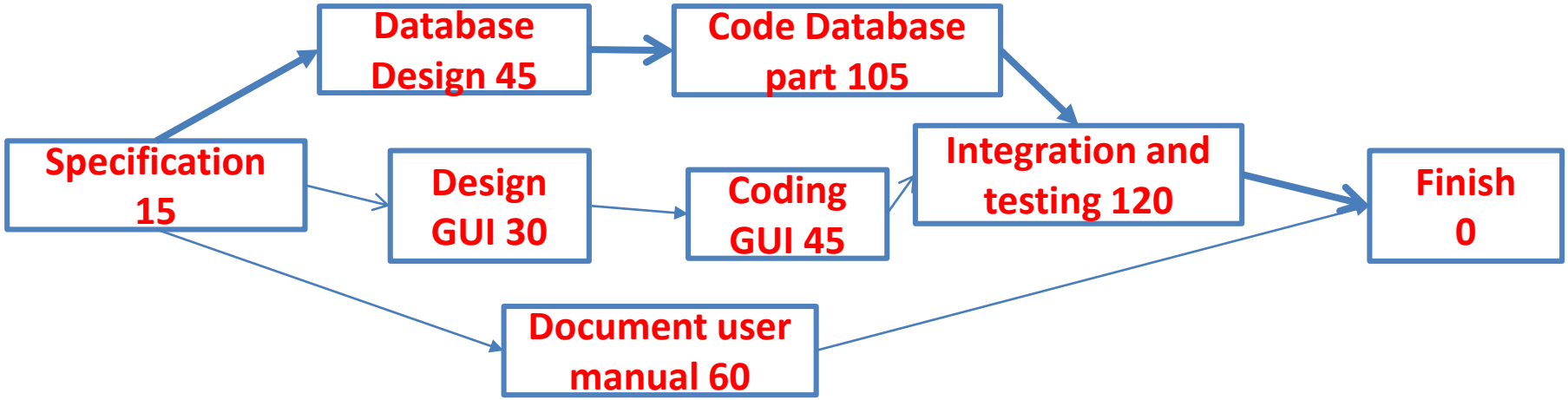


- The **latest start (LS) time** is the **difference between MT and the maximum of all paths from this task to the finish**.
  - For Example: **LS** for **specification** is: **285-max(285,210,75)=0**

Critical Path Method (CPM)

- The **latest start (LS) time** is the **difference between MT and the maximum of all paths from this task to the finish**.
  - For Example: **LS** for **specification** is: **285-max(285,210,75)=0**
  - **Another ex: LS** for **Design GUI** is: **285-max(195)=90**

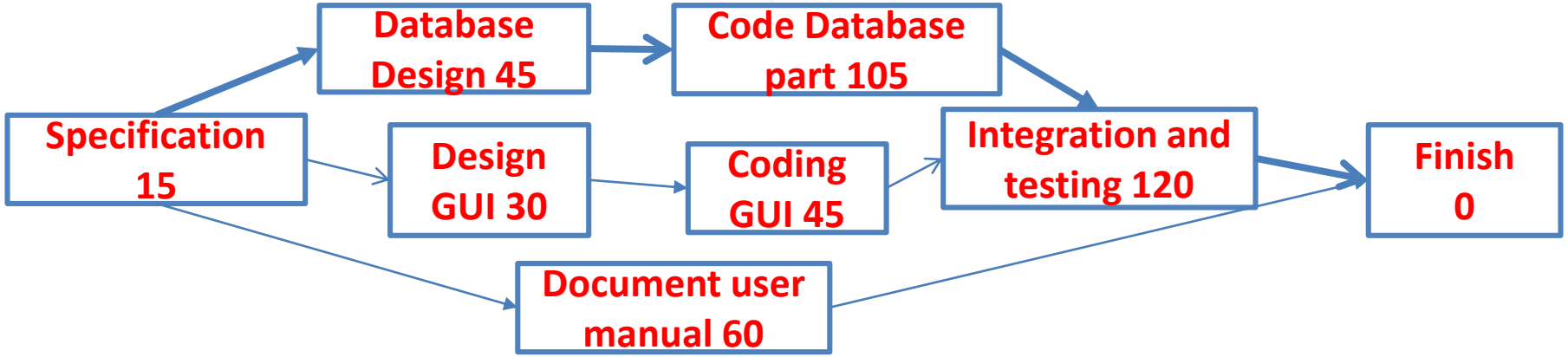
Task	Specification	Design database	Design GUI part	Code database	Code GUI part	Integrate and test	Write user manual
LS	0	15	90	60	120	165	225



Critical Path Method (CPM)

- The **latest finish (LF)** time of a task ?

Task	Specification	Design database	Design GUI part	Code database	Code GUI part	Integrate and test	Write user manual
LF	?	?	?	?	?	?	?

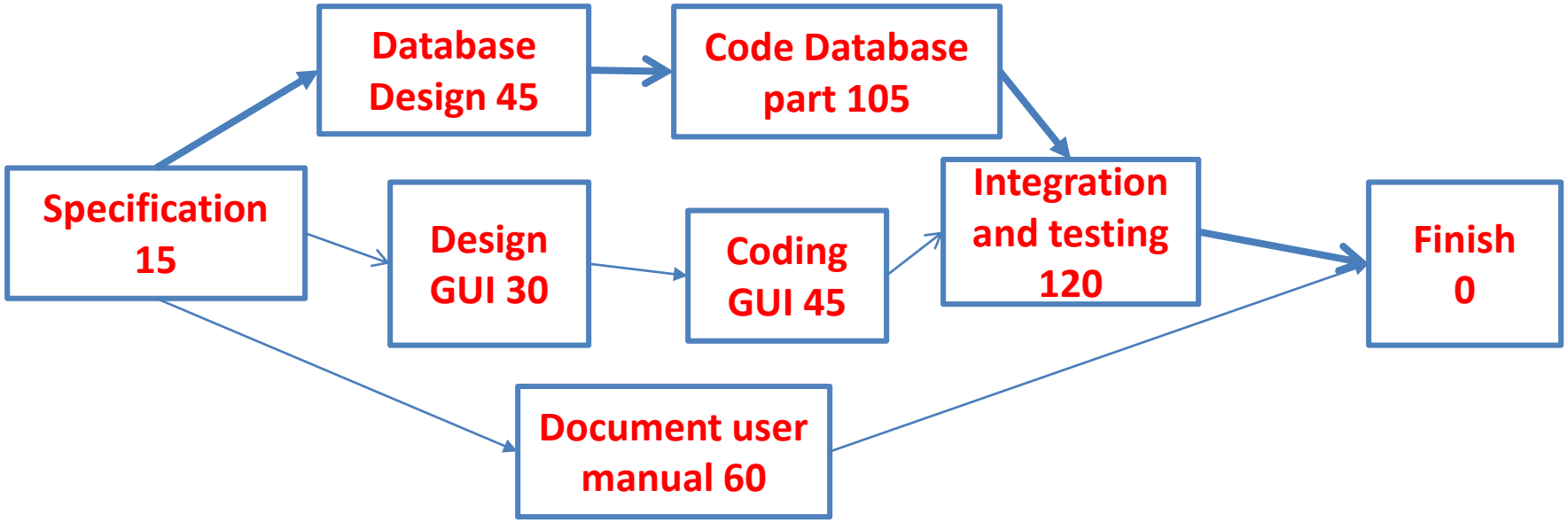


- The **latest finish (LF)** time of a task can be obtained by **subtracting maximum of all paths from this task to finish from MT** (alternatively, **LS + task duration**).

Critical Path Method (CPM)

- The **latest finish (LF)** time of a task can be obtained by **subtracting maximum of all paths from this task to finish from MT** (alternatively, **LS + task duration**).
  - For Example: **LF** for **specification** is: **285-(45+105+120)=15**
  - Another e.g.: **LF** for **Design GUI** is: **285-(45+120)=120**

Task	Specification	Design database	Design GUI part	Code database	Code GUI part	Integrate and test	Write user manual
LF	15	60	120	165	165	285	285



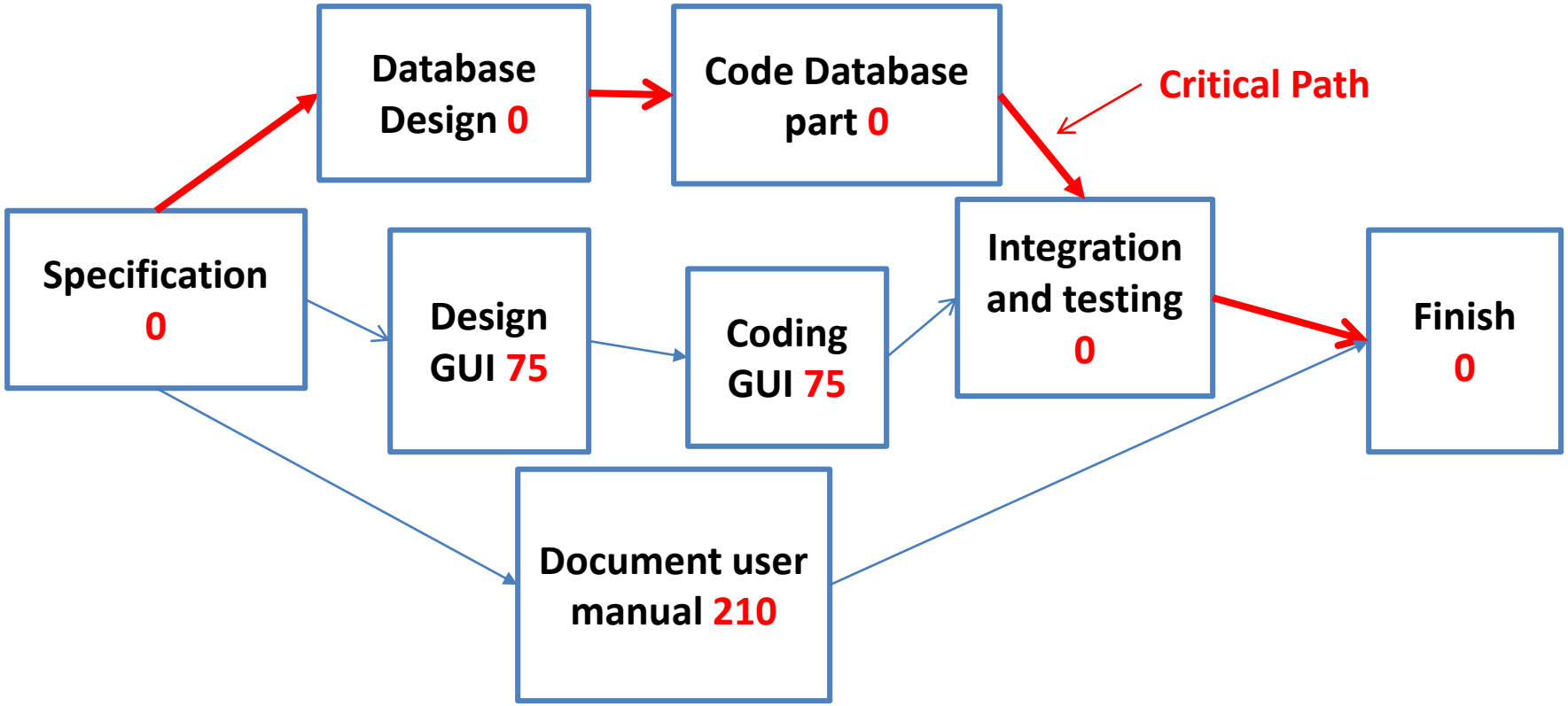
Critical Path Method (CPM)

- The **slack time (ST)** is **LS – ES** and **equivalently** can be written as **LF – EF**.
- The **slack time (or float time)** is **the total time that a task may be delayed before it will affect the end time of the project**. The slack time indicates the "flexibility" in starting and completion of tasks.

Time	Specification	Design database	Design GUI	Code database	Code GUI	Integrate and test	Write user manual
ES (earliest start)	0	15	15	60	45	165	15
LS (latest start)	0	15	90	60	120	165	225
EF (earliest finish)	15	60	45	165	90	285	75
LF (latest finish)	15	60	120	165	165	285	285
ST (Slack time)	**0**	**0**	**75**	**0**	**75**	**0**	**210**

Critical Path Method (CPM)

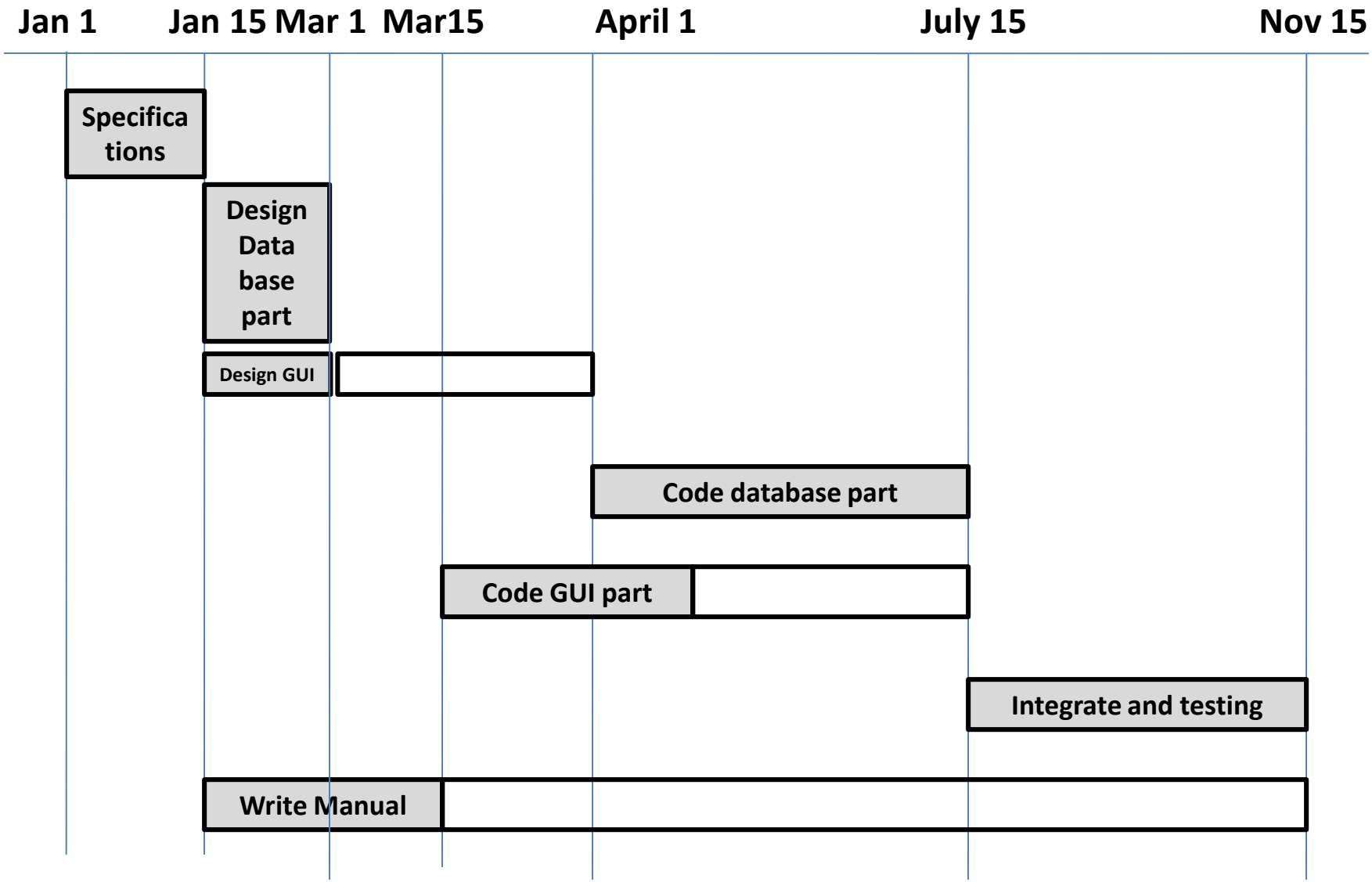
- **A critical task is one with a zero slack time (ST).**
- A **path from the start node to the finish node containing only critical tasks** is called **a critical path.**



Gantt chart (named after its developer Henry Gantt)

- Gantt charts are mainly used to present allocation of resources to the activities.
  - The resources allocated to activities include staff, hardware, and software.
  - **Gantt charts** are useful for **resource** planning.
  - A Gantt chart is a special type of bar chart where each bar represents an activity.
  - The bars **are** drawn along a time line.
  - The length of each bar is proportional to **the duration of** time planned for the corresponding activity.
- In the Gantt charts used for software project management, each bar consists of a white **part** and a shaded **part**.
  - The shaded part of the bar shows the **length of** time **each task is** estimated **to take**.
  - The white part shows the slack time, that is, the **latest time by which a task must be finished**.

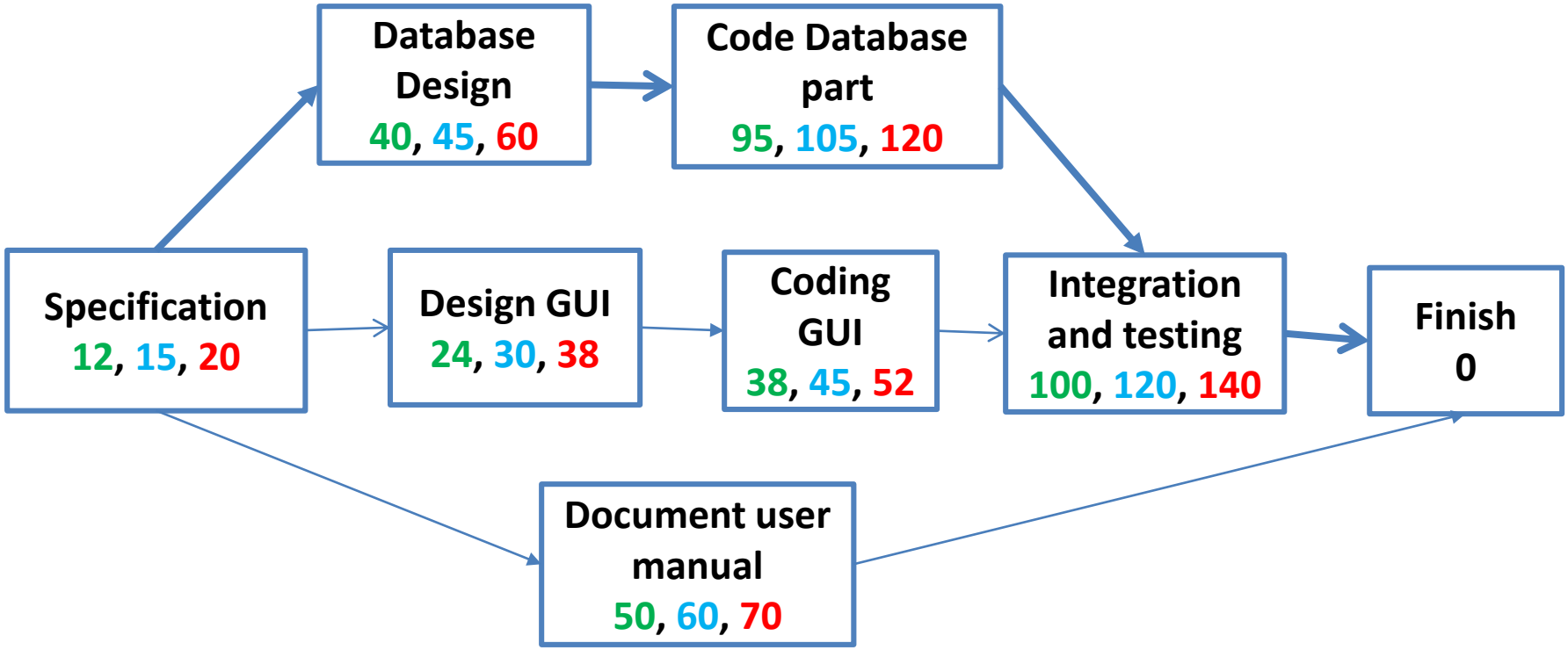
Gantt chart (named after its developer Henry Gantt)



Gantt chart representation of the MIS problem

PERT (Project Evaluation and Review Technique) chart

- This **chart consists of** a **network of boxes** and **arrows**.
  - The **boxes represent activities** and the **arrows represent task dependencies**.
  - **PERT chart represents** the **statistical variations in the project estimates** assuming a normal distribution.
  - In a PERT chart, **instead of making a single estimate** for each task, **pessimistic**, **likely**, and **optimistic estimates** are made. The boxes of PERT charts are usually annotated with the pessimistic, likely, and optimistic estimates for every task.
  - **Since all possible completion times between the minimum and maximum duration for every task has to be considered**, there are **not one but many critical paths**, depending on the permutations of the estimates for each task.
- A critical path in a PERT chart is shown by using thicker arrows.

PERT (Project Evaluation and Review Technique) chart

- **Gantt chart** representation of a project schedule **is helpful in planning the utilization of resources**, while **PERT chart is useful for monitoring the timely progress of activities**.
- Also, it is **easier to identify parallel activities** in a project using a PERT chart. Project managers need to identify the parallel activities in a project for assignment to different engineers.